

PCA of Residual Direction Vectors: Known (HP) vs Unknown (TOFU) Knowledge

$$\mathbf{d} = \text{mean}_{\text{batch}}(\mathbf{h}_{\text{reg}} - \mathbf{h}_{\text{anon}})$$

Qwen-1.5-72B (layer 17)

Qwen2.5-72B (layer 27)

Qwen2.5-14B (layer 47)

Gemma-2B (layer 18)

Qwen-7B (layer 0)

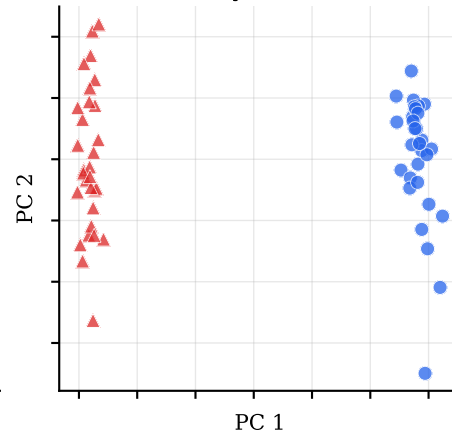
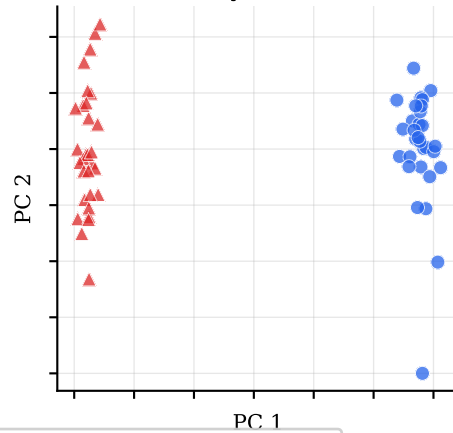
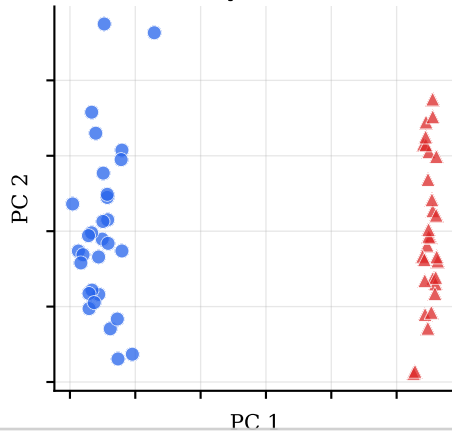
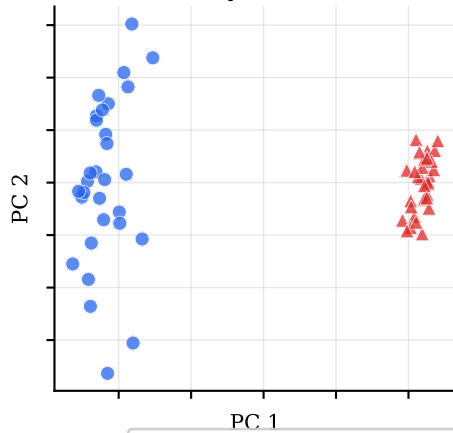
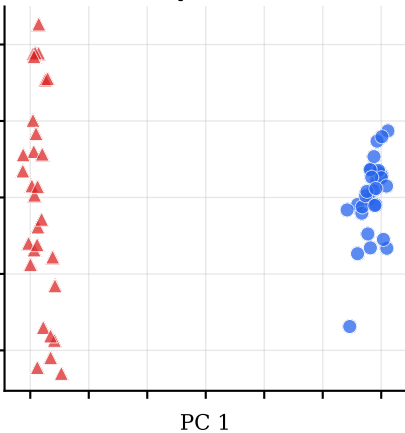
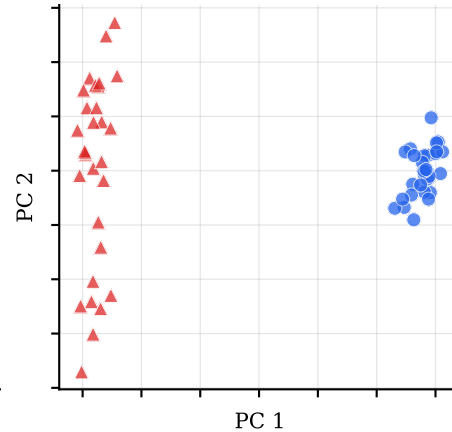
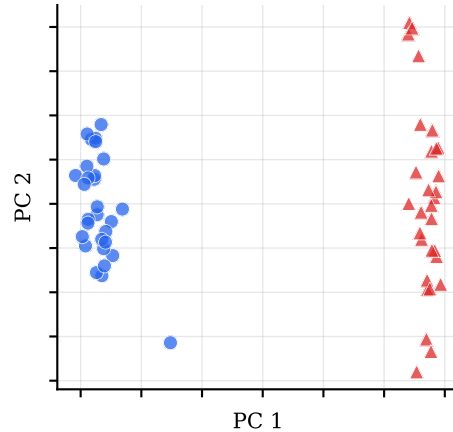
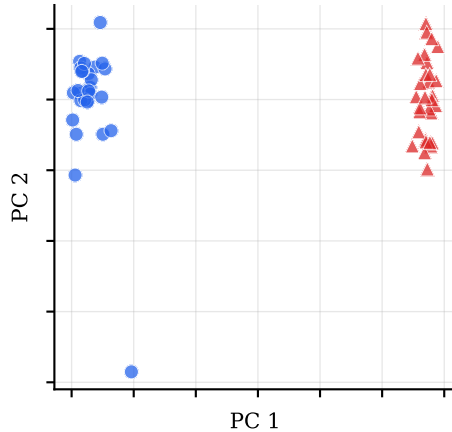
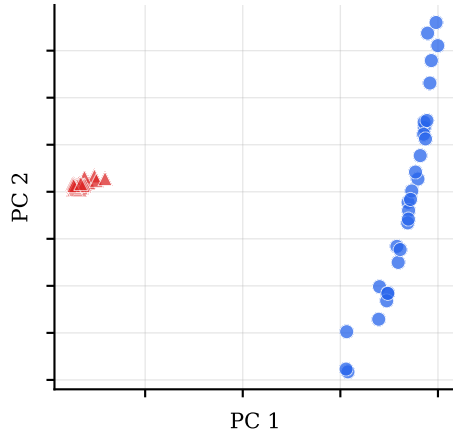
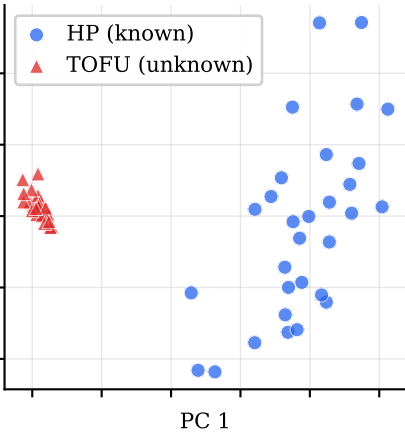
Gemma-7B (layer 26)

Llama-2-7B (layer 31)

Llama-2-13B (layer 28)

Llama-3-8B (layer 25)

Llama-3.1-8B (layer 25)



HP — Harry Potter (known domain) TOFU — Fictitious authors (unknown domain)